

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Pseudocode Writing Task

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO5 – Naturalization (independent application using OpenCV)P5	Assessment:	Assessment Tool 2 – Pseudocode Writing Task
Weightage:	25% of CIA	Total Marks:	100
CO4 Statement:	Develop computer vision applications using OpenCV for performing images and vid		
Student Name:	G. Tejasree	Reg. No.:	192511204
Section / Batch:	Section A	Date:	30 August 2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation	
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach	
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs	
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear	
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient	

8. Video Frame Extraction System

Problem Understanding:

The system reads a video, extracts frames at fixed intervals, and saves only the selected frames. It must use storage efficiently and avoid saving redundant frames. The assessment criteria emphasize accurate inputs/outputs, complete logic, appropriate loops and conditions, clear structure, and efficient handling of cases.

Pseudocode

BEGIN

1. INPUT video_file

2. INPUT interval

3. OPEN video_file

4. IF video cannot be opened THEN
 DISPLAY "Error: Video not found"
 STOP
END IF

5. INITIALIZE frame_count $\leftarrow$ 0

6. INITIALIZE saved_count $\leftarrow$ 0

7. CREATE output_folder

8. WHILE video has frames DO

 READ current_frame

 IF frame cannot be read THEN
 BREAK
 END IF

 frame_count $\leftarrow$ frame_count + 1

 IF frame_count MOD interval = 0 THEN

 IF current_frame is not empty THEN

 GENERATE filename using saved_count

 SAVE current_frame

 IF frame saved successfully THEN
 saved_count $\leftarrow$ saved_count + 1
 END IF

```

        END IF

    END IF

END WHILE

9. CLOSE video_file

10. DISPLAY "Total frames processed:", frame_count
11. DISPLAY "Frames saved:", saved_count
END

```

Modular Version

```

FUNCTION OPEN_VIDEO(video_file)
    Open video
    IF video is not opened
        RETURN failure
    RETURN video
END FUNCTION
FUNCTION SHOULD_SAVE(frame_count, interval)
    IF frame_count MOD interval = 0
        RETURN TRUE
    ELSE
        RETURN FALSE
    END FUNCTION

```

```

FUNCTION SAVE_FRAME(frame, saved_count)
    CREATE unique filename
    SAVE frame
    RETURN success
END FUNCTION

```

```

MAIN
    video ← OPEN_VIDEO(video_file)

    IF video failed
        DISPLAY error
        STOP
    END IF

    frame_count ← 0
    saved_count ← 0

```

```

WHILE video has frames
  READ frame
  frame_count ← frame_count + 1

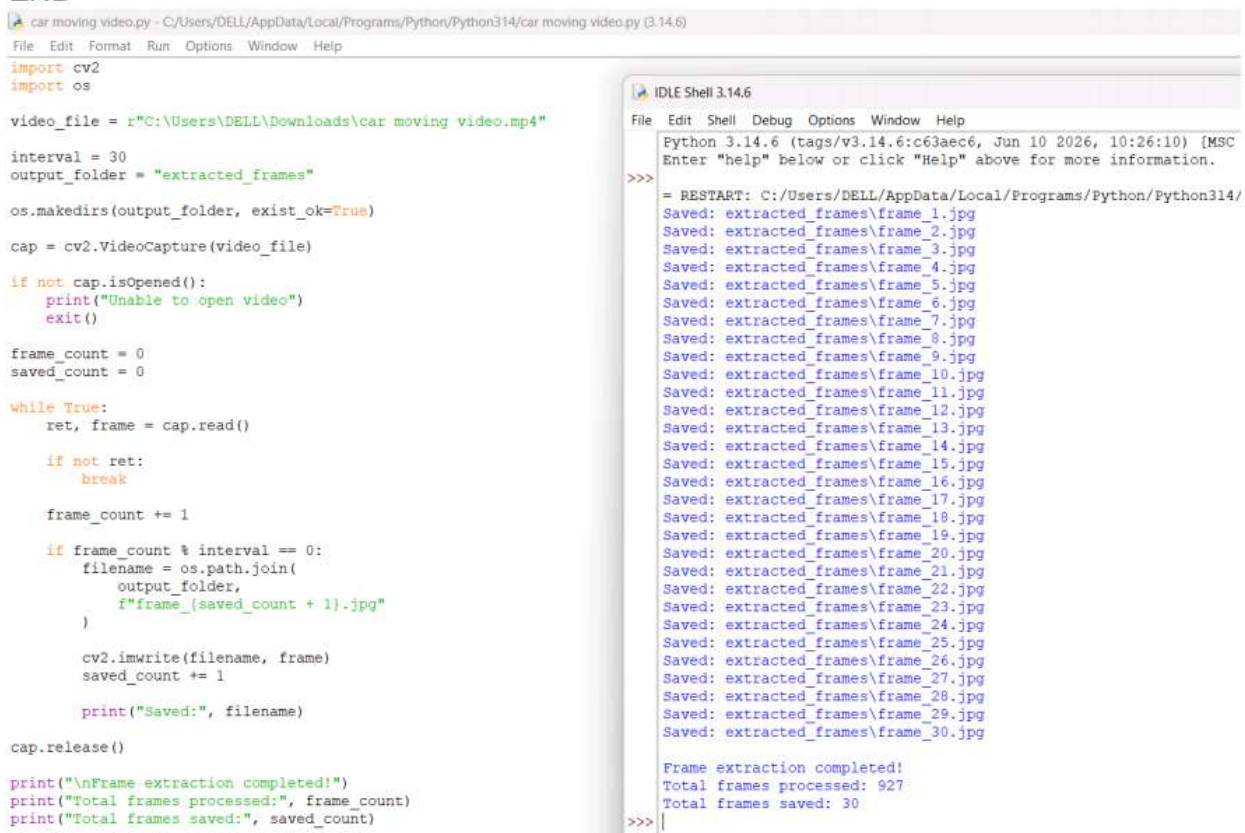
  IF SHOULD_SAVE(frame_count, interval) THEN
    IF SAVE_FRAME(frame, saved_count) THEN
      saved_count ← saved_count + 1
    END IF
  END IF
END IF
END WHILE

```

```

RELEASE video
DISPLAY processing summary
END

```



The screenshot shows a Python script named 'car moving video.py' in the IDLE editor. The script uses the cv2 library to read frames from a video file and save selected frames as JPEG images. The output window shows the execution results, including a list of saved frames and summary statistics.

```

car moving video.py - C:/Users/DELL/AppData/Local/Programs/Python/Python314/car moving video.py (3.14.6)
File Edit Format Run Options Window Help

import cv2
import os

video_file = r"C:\Users\DELL\Downloads\car moving video.mp4"

interval = 30
output_folder = "extracted_frames"

os.makedirs(output_folder, exist_ok=True)

cap = cv2.VideoCapture(video_file)

if not cap.isOpened():
    print("Unable to open video")
    exit()

frame_count = 0
saved_count = 0

while True:
    ret, frame = cap.read()

    if not ret:
        break

    frame_count += 1

    if frame_count % interval == 0:
        filename = os.path.join(
            output_folder,
            f"frame_{saved_count + 1}.jpg"
        )

        cv2.imwrite(filename, frame)
        saved_count += 1

        print("Saved:", filename)

cap.release()

print("\nFrame extraction completed!")
print("Total frames processed:", frame_count)
print("Total frames saved:", saved_count)

```

```

IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/DELL/AppData/Local/Programs/Python/Python314/
Saved: extracted_frames\frame 1.jpg
Saved: extracted_frames\frame 2.jpg
Saved: extracted_frames\frame 3.jpg
Saved: extracted_frames\frame 4.jpg
Saved: extracted_frames\frame 5.jpg
Saved: extracted_frames\frame 6.jpg
Saved: extracted_frames\frame 7.jpg
Saved: extracted_frames\frame 8.jpg
Saved: extracted_frames\frame 9.jpg
Saved: extracted_frames\frame 10.jpg
Saved: extracted_frames\frame 11.jpg
Saved: extracted_frames\frame 12.jpg
Saved: extracted_frames\frame 13.jpg
Saved: extracted_frames\frame 14.jpg
Saved: extracted_frames\frame 15.jpg
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Saved: extracted_frames\frame 23.jpg
Saved: extracted_frames\frame 24.jpg
Saved: extracted_frames\frame 25.jpg
Saved: extracted_frames\frame 26.jpg
Saved: extracted_frames\frame 27.jpg
Saved: extracted_frames\frame 28.jpg
Saved: extracted_frames\frame 29.jpg
Saved: extracted_frames\frame 30.jpg

Frame extraction completed!
Total frames processed: 927
Total frames saved: 30

>>>

```

Output: Selected frames are saved at the specified fixed interval, while unnecessary/redundant frames are not stored.